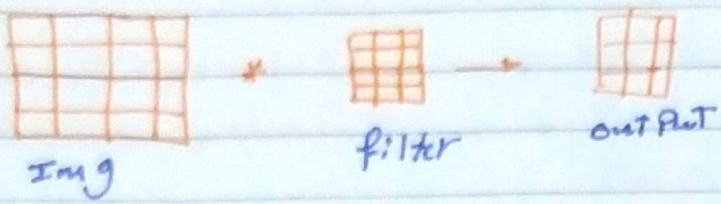
** In Image classification ANN has some limitations
- To much computation
- Treats local pixels same as pixels far apart
- Sensitive to location for the object in the image

Human filters images by extracting features from it and that is what we want the computer make using CNN

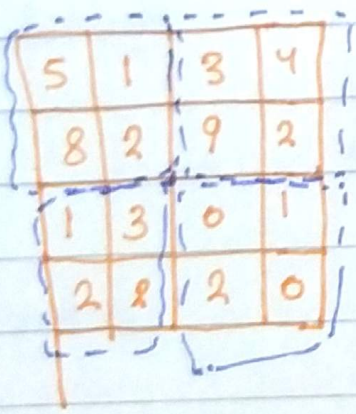
← Convolution: Applying matrix of numbers "filter" to an image



when appling convolution to the image it's activated "gives 1" when the feature existing in the image.

But we still have the problem of large size so we use pooling

we use pooling to reduce dimentions

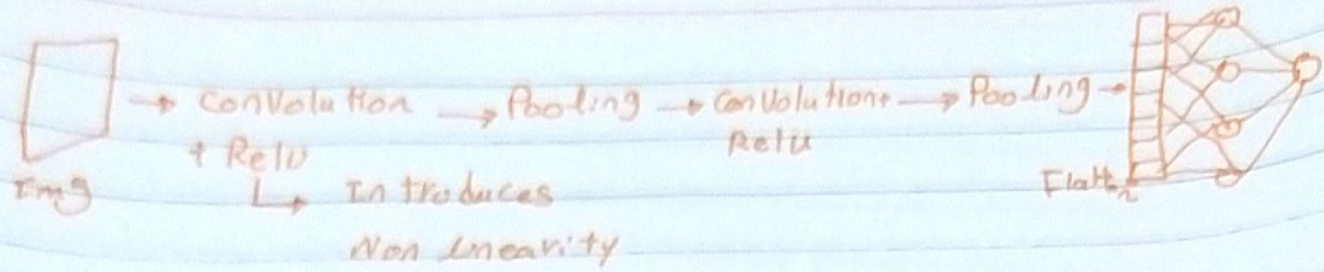


* Max Pooling →

8	9
3	2

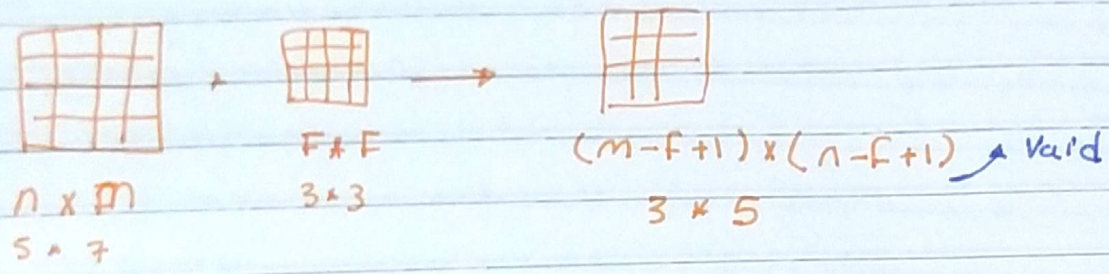
* Average Pooling →

4	4.5
2	.75

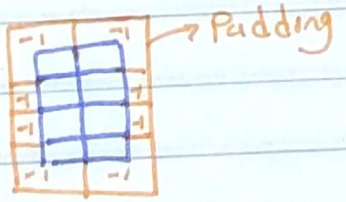


* CNN can't handel thickness & Rotation itself so we need To have rotated & skewed examples in our traing set
- or use data augmentation methods to generate new scaled & skewed samples from the existing training set

- Valid convolution ---> No Padding
- Same convolution --> Pad such as the output is same as the input



* The disadvantage of valid convolution is that the features in the corner don't play important role as in clasification so we use Padding



- stride عدد الـ pixels اللي بنتحركها يمين وتحت

(1,1) ... (2,2) ---